Real Time Improvement Research on Deep Learning Based Artificial Intelligence Computer Vision Image Dehazing Technology

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***Abstract*—Image dehazing technology is a crucial research direction in the field of Computer Vision (CV), whose core goal is to eliminate the impact of atmospheric scattering on image quality and restore clear scene information. Traditional dehazing algorithms, though simple in principle, lack robustness in complex foggy scenarios; AI dehazing methods based on deep learning can improve dehazing performance but are difficult to meet real-time requirements (e.g., autonomous driving, real-time monitoring) due to large model parameter size and high computational complexity. Based on the machine learning framework, this paper focuses on the real-time optimization of CV image dehazing. By designing a lightweight network structure, improving feature extraction strategies, and introducing model acceleration technologies, an AI dehazing model with both dehazing accuracy and real-time performance is constructed. Experiments are conducted based on the RESIDE dataset as the test benchmark. Compared with traditional methods and existing deep learning methods, the proposed model maintains a Peak Signal-to-Noise Ratio (PSNR) of 28.6 dB and a Structural Similarity Index (SSIM) of 0.91, while the inference speed is increased to 62 FPS (Frames Per Second), meeting the real-time processing requirements. This research provides an effective technical solution for image dehazing applications in real-time CV scenarios, and has important theoretical significance and engineering value.**

Keywords—Machine Learning; Computer Vision; Image Dehazing; Real-Time Optimization; Lightweight Network

I. INTRODUCTION

Adverse weather conditions such as fog and haze reduce image contrast and blur details, seriously affecting subsequent tasks of CV systems (e.g., object detection, semantic segmentation, scene recognition). In scenarios with high real-time requirements like autonomous driving, intelligent monitoring, and remote sensing detection, the balance between accuracy and speed of image dehazing technology has become a key bottleneck restricting system performance[1-6].

Traditional image dehazing methods are centered on physical models, such as the histogram equalization method proposed by Tan et al. and the Dark Channel Prior (DCP) algorithm proposed by He et al. These methods rely on manually designed features and prior assumptions, with low computational complexity but poor adaptability to heavy fog and uniform fog scenarios, and the dehazed images are prone to color distortion. With the development of artificial intelligence and machine learning, deep learning-based dehazing methods (e.g., DehazeNet, AOD-Net, FFA-Net) automatically learn dehazing features through end-to-end training, significantly improving dehazing accuracy. However, these models usually contain a large number of convolutional layers and parameters (e.g., FFA-Net has 11.2M parameters), resulting in slow inference speed (only 5-15 FPS), which cannot meet the needs of real-time scenarios.

Therefore, optimizing the real-time performance of AI CV image dehazing under the machine learning framework not only needs to break through the trade-off between "accuracy and speed" but also promotes the practical application of dehazing technology in engineering, which has important research value.

II. RELATED THEORETICAL FOUNDATIONS

A. Atmospheric Scattering Physical Model

As shown in Formula (1), the essence of image dehazing is to estimate the transmittance $t(x)$ and atmospheric light A from the foggy image $I(x)$, and then solve the haze-free image $J(x)$. By transforming Formula (1), the restoration formula for the haze-free image is obtained:

$$J(x) = \frac{I(x) - A(1 - t(x))}{t(x)} \quad (1)$$

When $t(x)$ is too small (heavy fog regions), the denominator approaches 0, which easily causes noise in $J(x)$. Therefore, a constraint is usually imposed on the transmittance:

$$t(x) = \max(t_0, t_{est}(x)) \quad (2)$$

Where $t0$ is the minimum transmittance (generally set to 0.1), and $t_{est}(x)$ is the estimated transmittance.

B. Fusion Logic Between Machine Learning Framework and CV Image Dehazing

Machine learning-based dehazing methods essentially replace manual priors with data-driven approaches, using neural networks to learn the mapping relationship from $I(x)$ to $J(x)$ or $t(x)$. The core process includes:

Data Preprocessing: Construct a paired dataset of foggy and haze-free images (e.g., RESIDE), and perform operations such as cropping and normalization on the images;

Model Training: Train the network to learn dehazing features using Mean Squared Error (MSE) or SSIM as the loss function;

Inference Optimization: Improve real-time performance through machine learning engineering technologies such as model compression, quantization, and parallel computing.

C. Key Technologies for Real-Time Optimization

Model quantization converts floating-point parameters (FP32) into low-precision integers (INT8), reducing memory usage and computational latency. Its core is to learn quantization parameters through a calibration set to ensure that the accuracy loss is within an acceptable range. The inference speed of the quantized model can usually be increased by 2-3 times.

III. Design of Real-Time Dehazing Model Based on Machine Learning

A. Overall Model Architecture

The Lightweight Dehaze Network (LDN) proposed in this paper takes "feature extraction - feature enhancement - transmittance estimation - image restoration" as the core process.

Input Layer: Receive a foggy image I(x) with a size of 3 \times 256 \times 256;

Lightweight Feature Extraction Module: Composed of 3 layers of depthwise separable convolution, outputting a 64-channel feature map;

Adaptive Channel Attention Module: Dynamically generate channel weights through the Sigmoid activation function to enhance effective features;

Fast Transmittance Estimation Module: Compress the feature dimension using 1×1 convolution, outputting the transmittance map $t_{est}(x)$;

Atmospheric Light Estimation Module: Quickly calculate the atmospheric light A based on the dark channel statistical method;

Image Restoration Module: Calculate the haze-free image J(x) by substituting into Formula (2);

Output Layer: Output a haze-free image with a size of 3 \times 256 \times 256.

B. Core Module Design

This module adopts a combined structure of depthwise separable convolution + Batch Normalization (BN) + ReLU activation, with specific parameters as follows:

Layer 1: Depthwise separable convolution (3×3, stride=1, padding=1), input channels=3, output channels=32, BN+ReLU;

Layer 2: Depthwise separable convolution (3×3, stride=1, padding=1), input channels=32, output channels=64, BN+ReLU;

Layer 3: Depthwise separable convolution (3×3, stride=1, padding=1), input channels=64, output channels=64, BN+ReLU.

Through depthwise separable convolution, the computational complexity of this module is only 1/8 of that of traditional convolution, and the number of parameters is controlled within 0.05M.

To solve the problem of insufficient feature expression capability of lightweight networks, a channel attention mechanism is introduced, and its calculation process is as follows:

Perform Global Average Pooling (GAP) on the feature map to obtain a channel-level feature vector $F_{gap} \in \mathbb{R}^{64}$;

Compress the dimension through a 2-layer Fully Connected (FC) network: F{fc1} = ReLU(FC(Fgap, 64 \to 16));

Restore the dimension and generate attention weights: W = Sigmoid(FC(F{fc1}, 16 \to 64));

Feature weighting:F{att} = F \times W (element-wise multiplication by channel), where F is the input feature map.

This module dynamically adjusts channel weights through machine learning, increasing the feature utilization rate while adding a small amount of computational complexity (with 0.01M parameters).

Traditional deep learning dehazing models require complex networks to estimate transmittance. This paper simplifies it into a lightweight structure of "1×1 convolution + Sigmoid activation":

1×1 convolution: Input channels=64, output channels=1, compressing the feature map into a single channel;

Sigmoid activation: Map the output value to the interval [0,1] to obtain the estimated transmittance value t_{est}(x);

Impose constraint: Substitute into Formula (3) to obtain the final transmittance t(x).

This module has only 0.02M parameters, and its computational speed is more than 5 times faster than that of traditional methods.

Lightweight technology description: The core lightweight design of the LDN model is based on depthwise separable convolution (replacing traditional standard convolution) and combined with the channel attention mechanism to achieve a balance between "efficient extraction and precise enhancement". Depthwise separable convolution decouples spatial convolution and channel convolution, and under the same receptive field, the computational complexity is only 1/8 of that of traditional convolution (specific calculation: Traditional convolution FLOPs = Input channels × Output channels × Kernel size^2 × Feature map size^2; Depthwise separable convolution FLOPs = Input channels × Kernel size^2 × Feature map size^2 + Input channels × Output channels × Feature map size^2); The channel attention mechanism dynamically adjusts channel weights through a 2-layer lightweight fully connected network, adding only 0.01M parameters, avoiding the computational overhead caused by complex attention structures.

Parameter/FLOP details:

Depthwise separable convolution module: Total parameters 0.05M, FLOPs 0.3G;

Adaptive channel attention module: Total parameters 0.01M, FLOPs 0.08G;

Fast transmittance estimation module: Total parameters 0.02M, FLOPs 0.12G;

Overall model: Total parameters 0.08M, total FLOPs 0.5G (corresponding to data in Table 2).

The novelty of this design lies in: on the basis of adopting mature lightweight technologies, through module-level structural simplification (such as simplifying transmittance estimation to "1×1 convolution + constraint") and functional collaboration (feature complementarity between attention mechanism and depthwise separable convolution), a better balance between parameter scale and performance is achieved. Compared with AOD-Net (0.1M parameters, 0.8G FLOPs), while reducing parameters by 20%, FLOPs are reduced by 37.5%.

C. *Model Training Strategy*

The RESIDE dataset, commonly used in the field of CV image dehazing, is adopted, including the training set ITS (Indoor Training Set, 13,990 pairs of images) and the test set OTS (Outdoor Test Set, 500 pairs of images). Preprocessing steps:

Image Cropping: Crop the original images into a size of 256 \times 256 to avoid overfitting during training;

Data Augmentation: Randomly flip and rotate the images to improve the generalization ability of the model;

Normalization: Normalize the pixel values from [0,255] to [0,1].

A hybrid loss function of "MSE + SSIM" is adopted to balance dehazing accuracy and image structure preservation:

$$Loss = \alpha \cdot MSE(J_{pred}, J_{gt}) + (1-\alpha) \cdot (1 - SSIM(J_{pred}, J_{gt})) \quad (3)$$

Where Jpred is the haze-free image output by the model, Jgt is the real haze-free image, and α is the weight coefficient (set to 0.3).

Framework: PyTorch 1.10;

Optimizer: Adam, initial learning rate=0.001, decaying to 0.5 of the original every 10 epochs;

Batch Size: 32;

Training Epochs: 50;

Hardware: NVIDIA RTX 3090 GPU.

D. *Model Real-Time Optimization Scheme*

The trained FP32 model is quantized into an INT8 model based on the TensorRT framework, with steps as follows:

Select 100 images from the ITS dataset as the calibration set;

Adopt a quantization strategy of minimizing KL divergence to calibrate the distribution difference between the floating-point and integer models;

Generate an INT8 engine file for inference acceleration.

Use the CUDA cores of the GPU to realize parallel computing of feature extraction and transmittance estimation. At the same time, reduce memory access latency through the layer fusion technology of TensorRT (e.g., merging convolution, BN, and ReLU into a single operation).

IV. Experimental Results and Analysis

A. *Experimental Environment and Evaluation Indicators*

Hardware: Intel i9-12900K CPU, NVIDIA RTX 3090 GPU (24GB), 32GB DDR4 memory;

Software: Ubuntu 20.04, PyTorch 1.10, TensorRT 8.4, OpenCV 4.5;

Test Image Size: 512 \times 512 (simulating the image resolution in actual scenarios).

Dehazing Accuracy: PSNR (the higher the better) and SSIM (the closer to 1 the better), reflecting the difference between the haze-free image and the real image;

Real-Time Performance: FPS (the higher the better) and inference time (processing time per image, the lower the better), reflecting the speed performance of the model;

Model Complexity: Number of parameters (Params, the smaller the better) and computational complexity (FLOPs, the smaller the better), reflecting the lightweight degree of the model.

B. *Comparative Experiment Design*

Five types of representative methods are selected for comparison:

Traditional methods: DCP (Dark Channel Prior), GF (DCP optimized by Guided Filtering);

Lightweight deep learning methods: AOD-Net, DehazeNet;

High-accuracy deep learning methods: FFA-Net, TransHaze;

Proposed methods: LDN (unquantized), LDN-TensorRT (quantized).

C. *Experimental Result Analysis*

The accuracy results on the RESIDE-OTS test set are shown in Table 1:

Table 1 Accuracy on the RESIDE-OTS test set

Method	PSNR (dB)	SSIM
DCP	18.2	0.65
GF	19.5	0.69
AOD-Net	22.3	0.78
DehazeNet	23.1	0.80
FFA-Net	30.5	0.94
TransHaze	31.2	0.9

As can be seen from Table 1:

A comprehensive performance comparison reveals distinct gaps between traditional and deep learning-based dehazing approaches. Traditional methods like Dark Channel Prior (DCP) and Guided Filter (GF) show significantly lower Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) compared to deep learning models. This disparity stems from the former's reliance on handcrafted features and empirical assumptions, which fail to adapt to complex real-world haze variations. In contrast, machine learning-driven models leverage data-driven feature learning to capture intricate haze patterns, directly validating their superior accuracy in dehazing tasks.

Regarding the proposed LDN model, its unquantized version exhibits marginally lower accuracy than high-performance counterparts such as FFA-Net, which benefit from complex architectures like multi-scale fusion and attention mechanisms. However, it outperforms lightweight models (AOD-Net, DehazeNet) by a substantial margin, as the latter prioritize computational efficiency over feature extraction capability, leading to compromised performance.

Notably, the quantized LDN-TensorRT model achieves a slight accuracy improvement, with PSNR rising from 27.8 dB to 28.6 dB. This enhancement is attributed to the calibration strategy employed during quantization, which optimizes the statistical distribution of feature maps. By adjusting key parameters to preserve critical feature information, the strategy mitigates the potential precision loss of quantization, enabling the model to maintain efficiency while boosting dehazing accuracy.

The results of real-time performance and complexity are shown in Table 2:

Table 2 Results of Real time Performance and Complexity

Method	Params (M)	FLOPs (G)	Inference Time (ms/image)	FPS
DCP	0.001	0.1	33.2	30.1
GF	0.002	0.3	45.5	22.0
AOD-Net	0.1	0.8	22.1	45.2
DehazeNet	0.5	1.2	38.5	26.0
FFA-Net	11.2	25.6	200.1	5.0
TransHaze	20.5	42.8	125.3	8.0
LDN (Unquantized)	0.08	0.5	18.3	54.6
LDN-TensorRT	0.08 (INT8)	0.5	16.1	62.1

As can be seen from Table 2:

Traditional methods have the smallest number of parameters, but their real-time performance (30.1 FPS) is still lower than that of the method in this paper, and their accuracy is insufficient;

High-accuracy methods (FFA-Net, TransHaze) have huge parameters and computational complexity, with FPS only 5-8, which cannot meet real-time requirements;

The proposed LDN model (unquantized) has only 0.08M parameters and 0.5G FLOPs, with FPS reaching 54.6, which is better than most lightweight methods;

The quantized LDN-TensorRT model maintains the same complexity, while the FPS is increased to 62.1, and the inference time is shortened to 16.1 ms/image, meeting the requirements of real-time CV scenarios (usually requiring ≥30 FPS).

Three typical foggy images (light fog, moderate fog, and heavy fog) from the RESIDE-OTS test set are selected for subjective comparison, and the results are described as follows:

Light fog scenario: The dehazed image by DCP is relatively dark, AOD-Net has slight color distortion, while the image restored by the proposed method has moderate brightness and clear details;

Moderate fog scenario: GF method produces edge artifacts, DehazeNet fails to completely remove fog, while the proposed method can effectively eliminate fog and retain details of buildings, trees, etc.;

Heavy fog scenario: FFA-Net has the best dehazing effect but slow inference speed. Although the proposed method is slightly inferior to FFA-Net, it can already meet visual requirements and has significant speed advantages.

To comprehensively assess the visual fidelity of the model (especially for safety-critical scenarios such as autonomous driving), supplementary no-reference image quality assessment metrics, NIQE (Natural Image Quality Evaluator) and BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator), are used to test images from the RESIDE-OTS test set. No-reference metrics do not require real fog-free images as a benchmark and can directly reflect the consistency between dehazed images and naturally clear images. The results are shown in the following table 3:

Table 3 No reference image quality reference indicators

Method	NIQE (lower better)	BRISQUE (lower better)
DCP	8.21	65.3
GF	7.85	62.1
AOD-Net	6.32	58.5
DehazeNet	5.98	56.2
FFA-Net	3.15	32.8
TransHaze	2.98	30.5
LDN (Unquantized)	4.26	41.3
LDN-TensorRT	4.02	39.8

As can be seen from the table, the performance of the LDN series models in no-reference metrics is superior to traditional methods and lightweight models but slightly inferior to high-precision models such as FFA-Net. This is basically consistent with the trend of full-reference metrics (PSNR/SSIM), verifying the competitiveness of the model in terms of visual naturalness. Meanwhile, the quantized LDN-TensorRT model remains stable in no-reference metrics, without visual quality degradation due to precision compression.

Typical scenes from the RESIDE-OTS test set (including scenes prone to artifacts such as strong light sources, complex textures, and edge details) are selected to compare the performance of various methods in dehazed results in terms of halo artifacts, color distortion, and fine texture retention. The specific analysis is as follows:

Halo artifacts: DCP and GF methods show obvious halo diffusion in strong light source areas (such as street lamps and car window reflections); AOD-Net and DehazeNet have slight edge halos; FFA-Net and LDN series models can effectively suppress halos. Among them, the LDN-TensorRT model has better control over artifacts at light source boundaries due to the quantization calibration strategy.

Color distortion: DCP dehazed images are generally dark, and GF has local color cast; DehazeNet is insufficient in restoring the blue color of the sky area; The LDN series models adjust the weight of color channels through the adaptive channel attention mechanism, and the color tone deviation of dehazed images from real fog-free images is less than 5%, with better color consistency than lightweight comparison models.

Fine texture retention: Traditional methods have blurring in details such as leaves and building textures; FFA-Net has the clearest texture restoration but is limited in inference speed; The LDN series models can retain more than 85% of fine textures (verified by edge detection algorithms), achieving a balance between texture fidelity and real-time performance under a lightweight architecture.

D. *Ablation Experiments*

To verify the effectiveness of each core module, ablation experiments are designed, and the results are shown in Table 4:

Table 4 Results of ablation test

Model Configuration	PSNR (dB)	SSIM	FPS
Basic Model (Without Attention)	25.3	0.85	56.2
Basic Model + Attention Module	27.8	0.89	54.6
Basic Model + Attention + Quantization	28.6	0.91	62.1

As can be seen from Table 4:

After adding the adaptive channel attention module, PSNR increases by 2.5 dB and SSIM increases by 0.04, which proves that the attention mechanism can effectively enhance feature expression;

After adding model quantization, PSNR further increases by 0.8 dB and FPS increases by 7.5, verifying the effectiveness of the quantization strategy.

V. Conclusion and Future Work

A. *Research Conclusion*

Based on the machine learning framework, this paper conducts research on the real-time issue of CV image dehazing. By designing a lightweight network structure, introducing an adaptive channel attention mechanism, and adopting model quantization acceleration technology, an LDN real-time dehazing model is constructed. The experimental results show that:

The proposed model achieves a PSNR of 28.6 dB and an SSIM of 0.91 on the RESIDE dataset, and its dehazing accuracy is better than traditional methods and lightweight deep learning methods;

The model has only 0.08M parameters, and the quantized inference speed reaches 62.1 FPS, meeting the real-time requirements of scenarios such as autonomous driving and real-time monitoring;

Ablation experiments verify the improvement effect of the attention module and quantization strategy on model performance, providing a feasible optimization path for real-time image dehazing technology.

B. *Future Work*

Further optimize the network structure, explore the fusion scheme of Transformer and lightweight convolution, and improve the dehazing accuracy in heavy fog scenarios while maintaining real-time performance;

Expand the dataset to actual vehicle-mounted and monitoring scenarios to enhance the engineering adaptability of the model;

Study the dehazing method based on multi-modal data fusion (e.g., infrared + visible light) to improve robustness under severe weather conditions.

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